Question 1

Source — Dest

Data link rate - 100 Mbps
RRT - 100 ms = 1×10^{-1} s
Frame size - 1000 bytes

a) $RTT = \quad + \left(\frac{m}{b}\right)$

d - propagation delay | m - m bit data frame | b - data rate of bps

$b = m / (RTT)$

$= \frac{1000 \times 8 \text{ bits}}{(1 \times 10^{-1} \text{ s})}$

$= \frac{10^3 \times 8 \text{ bits}}{10^{-1} \text{ s}}$

$= 8 \times 10^4 \text{ bits/s}$

$= 0.08 \text{ Mbs}$

b) RATE = Window size × b

Win size $W' = 10 \times 0.08$ Mbs
$= 0.8$ Mbs

Win size $1000 = 1000 \times 0.08$ Mbs
$= 80$ Mbs

c) Max win size = how many frames can I transmit in RTT

$= \frac{RTT \times \text{Data Rate}}{\text{Frame size}}$

$$= \frac{0.1s \times 10 \times 10^6 \text{ bps}}{1000 \times 8 \text{ b}}$$

$= 1250$ window size

d Any unexpected delays
Lost Packets

Question 2

```
                 10010100 11
11011 | 1011101010100000
      -  11011
          11000
          11011
           111
           00
           11101
           00001
           11101
          -11011
           11001
           0000
           11001
          -11011
            1000
            0000
            10000
           -11011
             10110
             11011
              1101 } Remainder
```

Question 3

Propagation Speed = 3×10^8 m/s

(a) distance = 30 m
link data rate = 100 Mbps
= 10^8 bps
Frame size = 1 Kbit
10^3 bits

$$\text{Propagation delay} = \frac{\text{Distance}}{\text{Signal speed}}$$
$$= \frac{30 \text{ m}}{3 \times 10^8 \text{ m/s}}$$
$$= 10^{-7} \text{ s}$$

$$\text{Transmission time} = \frac{\text{Packet size (b)}}{\text{Link Data rate (b/s)}}$$
$$= \frac{10^3 \text{ b}}{10^8 \text{ b/s}}$$
$$= 10^{-5} \text{ s}$$

$$\text{ratio} = \frac{10^{-7} \text{ s}}{10^{-5} \text{ s}} = 0.01$$

Here CSMA is more benefical since propagation delay is much smaller than transmission time nodes can sense for collisions effectively

Propagation speed = 3×10^8 m/s

b Distance = 300 m

Data link rate = 10 Gbps

$= 10 \times 10^9$ bps

$= 10^{10}$ bps

Frame size = 4 K bit

$= 4 \times 10^3$ bit

Propagation delay $= \frac{\text{Distance (m)}}{\text{Propagation speed (m/s)}}$

$= \frac{3 \times 10^2 \text{ m}}{3 \times 10^8 \text{ m/s}}$

$= 10^{-6}$ s

Transmission delay $= \frac{\text{Frame size (bit)}}{\text{Data link rate (bps)}}$

$= \frac{4 \times 10^3 \text{ b}}{10^{10} \text{ bps}}$

$= 4 \times 10^{-7}$ s

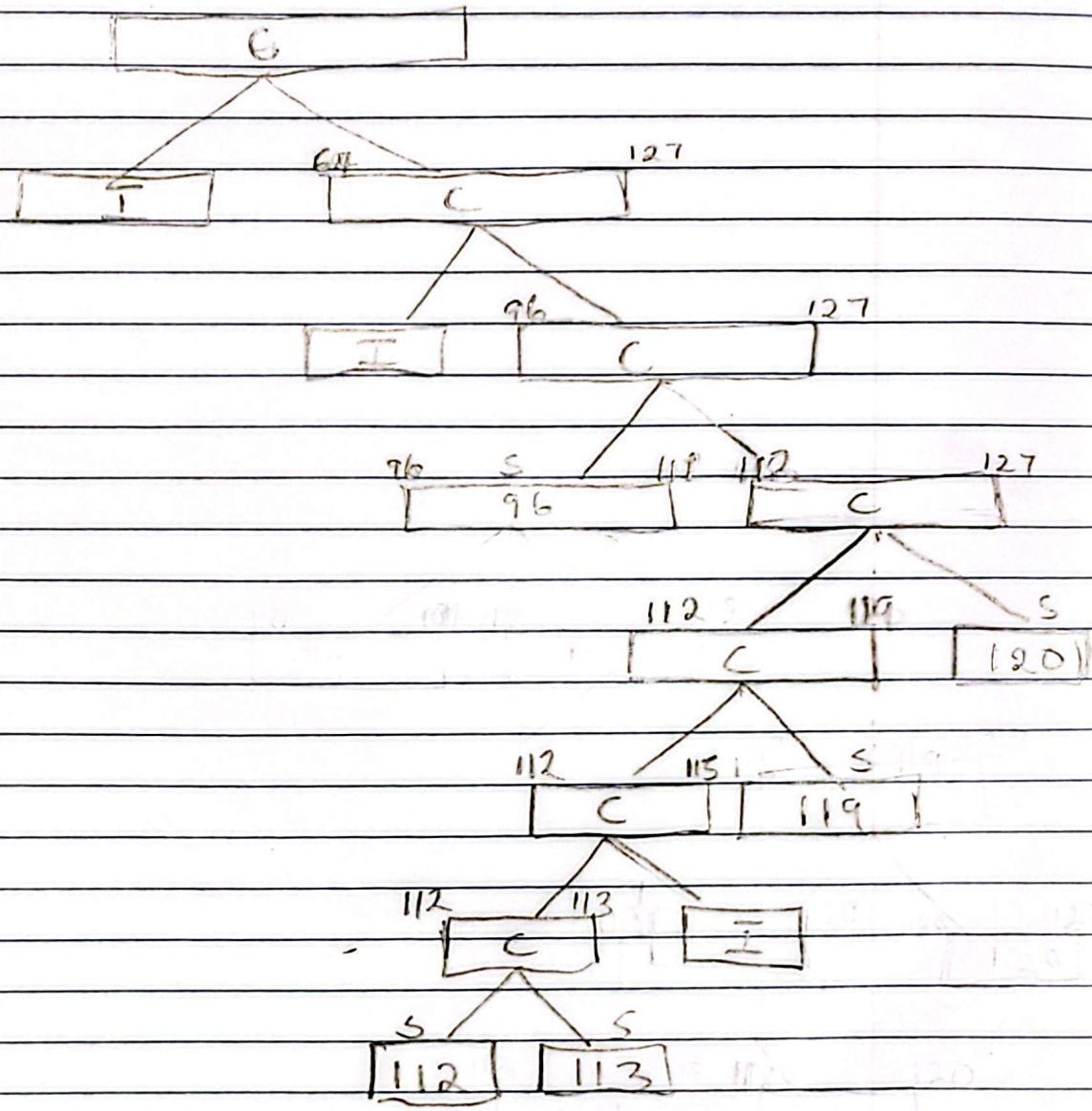
Ratio $= \frac{10^{-6} \text{ s}}{4 \times 10^{-7}}$

$= 0.25 \times 10$

$= 2.5$

In this scenario the propagation delay is higher than transmission time therefore ALOHA is more appropriate

ID's 96, 112, 113, 119, 120



a Time slot for each node

$2(N) - 1 = 2(128) - 1$

$= 255$

Total Time $= 255 \times 100\ \mu s = 25500\ \mu s$

b Only 15 active Nodes

$15 \times 100\ \mu s = 1500\ \mu s$